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Aerosol thermodynamics: problem description

Gibbs free energy minimization is a **non-linear constrained optimization problem**. Specifically, the *objective function* which we minimize (Gibbs free energy) is non-linear and non-convex with composition due to the non-linear activity coefficients γ_i. Following Wexler and Clegg (2002),

$$\text{minimize} \quad G = \sum_i n_i \mu_i \tag{1}$$

$$\text{subject to} \quad \sum_i \nu_i n_i = b, \tag{2}$$

$$n_i \geq 0 \tag{3}$$

Fortunately, the constraints (mass conservation and non-negativity), are linear. The chemical potential of each species i is

$$\mu_i = \mu_0(T, p) + RT \ln \gamma_i x_i, \tag{4}$$

where the activity coefficient γ_i is a non-linear function of the multicomponent chemical

References I

Wexler, A. S., & Clegg, S. L. (2002). Atmospheric aerosol models for systems including the ions h+, nh4+, na+, so42-, no3-, cl-, br-, and h2o. *Journal of Geophysical Research: Atmospheres*, *107*(D14), ACH–14.